

Critical Values for the KS, CvM, and AD Goodness-of-Fit Tests for the Log-Normal Distribution with Estimated Parameters

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Abstract

Standard critical values for EDF goodness-of-fit tests are invalid when parameters are estimated. This study examines the KS, CvM, and AD tests for a log-normal distribution with parameters estimated on the log scale, using Monte Carlo simulation to generate finite-sample critical values. Because $\ln X$ is normal, the fitted CDF values and all three test statistics are identical to those from the corresponding normality tests when the same estimators are used. The simulations confirm this equivalence within Monte Carlo variation. Thus, fully specified tables produce overly conservative tests, while normal estimated-parameter critical values can be used directly. This result follows from the logarithmic transformation and should not be generalized to unrelated distribution families.

1. INTRODUCTION

The fundamental task in a goodness-of-fit analysis is to test the null hypothesis (H_0) that a given random sample, $x_1, x_2, \dots, x_n$, originates from a population of a specific distribution function, $F(x)$. While several methods exist for this purpose, this paper focuses on a class of goodness-of-fit statistics known as *Empirical Distribution Function (EDF) statistics*. These tests are founded on the comparison between the hypothesized distribution function, $F(x)$, and the empirical distribution function, $F_n(x)$, derived from the sample data. Within the broad family of EDF statistics, this study will focus on three widely recognized and crucial tests: the **Kolmogorov-Smirnov (KS)**, the **Cramér-von Mises (CvM)**, and the **Anderson-Darling (AD)** tests.

For continuous distributions, EDF tests often provide greater power than traditional chi-square (χ^2) test while avoiding the need to group observations into bins. However, when distribution parameters are unknown and estimated from the same sample, standard EDF critical values are no longer valid. This estimated-parameter problem has historically limited the straightforward application of EDF tests.

This study focuses on generating critical value tables for the KS, AD, and CvM test statistics under a *log-normal* null hypothesis with *estimated parameters*. The paper is structured as follows: Section 2 provides a background on the EDF statistics. Section 3 details the Monte Carlo simulation procedure used to generate the critical

values. Section 4 presents the resulting tables and a validation of the findings against established benchmarks in literature. Finally, Section 5 offers concluding remarks.

1.1. THE LOG-NORMAL DISTRIBUTION

As noted above, the distribution selected for this investigation is the log-normal distribution. A continuous random variable $X \sim \text{LogNormal}(\mu, \sigma)$ is log-normally distributed if $Y = \ln(X)$ follows a normal distribution with mean μ and standard deviation σ . This log-normal distribution is widely used to model *non-negative, right-skewed, tail-heavy* data that arise in fields including finance, economics, engineering, and the biological sciences (Mathworks, n.d., Wolfram, n.d.).

(Wolfram, n.d.) The probability density function (PDF) of the log-normal distribution is given by:

$$f(x | \mu, \sigma) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln(x) - \mu)^2}{2\sigma^2}\right), \quad \text{for } x > 0$$

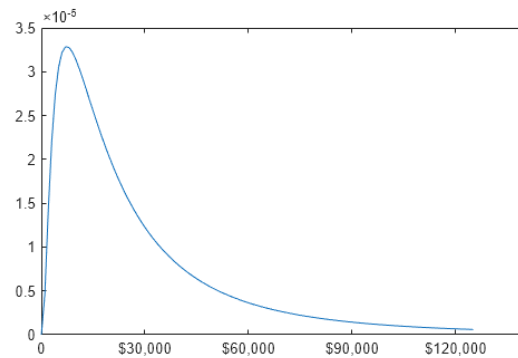


Figure 1. Example of a log-normal probability density function, illustrating its characteristic right skew (MathWorks, n.d.).

where μ and σ are the mean and standard deviation of $\ln X$ respectively.

The cumulative distribution function (CDF) can be expressed using the standard normal CDF, Φ :

$$F(x|\mu, \sigma) = \Phi\left(\frac{\ln(x) - \mu}{\sigma}\right)$$

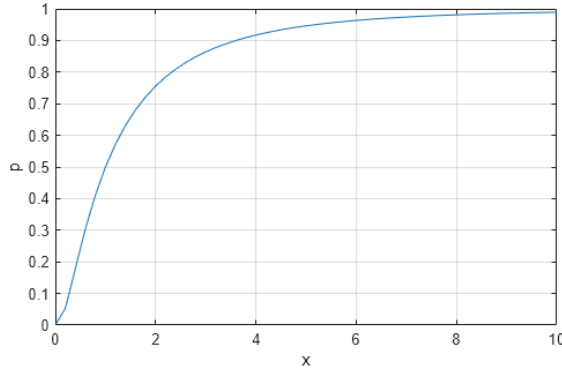


Figure 2. Example of a log-normal cumulative distribution function (MathWorks, n.d.).

The log-normal distribution has support on $(0, \infty)$, with zero as its lower bound, and is characterized by right skewness and a long right tail. This visual characteristic is confirmed by its theoretical skewness:

$$\gamma_1 = \sqrt{e^{\sigma^2} - 1} (2 + e^{\sigma^2})$$

For $(X \sim \text{LogNormal}(0,1))$ the skewness is approximately 6.183, whereas every normal distribution has zero skewness. Log-normal skewness is positive for every $\sigma > 0$ and increases with σ , which controls both the degree of skewness and the weight of the right tail.

The log-normal distribution was selected for this study by the following key factors:

- Although it is widely applied, comprehensive critical-value tables for EDF tests with estimated log-normal parameters are less readily available than those for distributions such as the normal distribution.
- Its logarithmic relationship to the normal distribution provides a direct theoretical benchmark for validating the simulation results.

Its skewness and long right tail may also be important when studying test power, particularly for the AD test. The present study, however, focuses on null critical values rather than power under alternative distributions.

2. BACKGROUND ON EDF GOODNESS-OF-FIT STATISTICS

2.1. THE KOLMOGOROV-SMIRNOV (KS) TEST

The Kolmogorov-Smirnov (KS) test measures the maximum vertical difference between the hypothesized cumulative distribution function (CDF), $F^*(x)$, and the empirical distribution function (EDF), $F_n(x)$, derived from the sample data. The test statistic, D , is formally defined as the supremum of this difference across all values of x :

$$D = \sup_x |F^*(x) - F_n(x)|$$

For practical computation from a set of ordered data points $(x_1 < x_2 < \dots < x_n)$, the statistic, denoted D_n , is calculated using the following components:

$$\begin{aligned} D_n^+ &= \max_i [i/n - F^*(x_i)] \\ D_n^- &= \max_i [F^*(x_i) - (i-1)/n] \\ D_n &= \max(D_n^+, D_n^-) \end{aligned}$$

LILLIEFORS ADAPTATION:

A critical issue with this standard procedure arises when parameters are unknown, a problem famously addressed by Lilliefors (1967). Using the standard tables in this scenario produces what Lilliefors describes as an "*extremely conservative*" test, meaning the actual probability of a TYPE I error is in fact significantly smaller than the level selected. To address this for the normal distribution, Lilliefors proposed a modification that has become standard practice. He constructed a new table of critical values specifically for the case where the mean and variance are estimated. The procedure retains the original statistic D , but with changes made in the definition of the hypothesized distribution, $F^*(x)$, where μ and σ^2 are replaced by their *sample* estimates. Using Monte Carlo simulation, Lilliefors generated the appropriate critical values for this statistic and found them to be substantially lower—approximately **two-thirds** of the standard KS values. This confirms that a different table of critical values is essential for an accurate test.

2.2. THE CRAMÉR-VON MISES (CVM) TEST

An alternative to the KS test is the Cramér-von Mises (CvM) statistic, W^2 , which evaluates the overall fit by considering squared difference between the empirical and hypothesized

distributions, rather than just the single point of maximum distance. The formal definition of the statistic is:

$$W_n^2 = n \int_{-\infty}^{\infty} [F_n(x) - F^*(x)]^2 dF^*(x)$$

For computational purposes, Stephens (1974) provides a more direct formula based on the transformed data points, $z_i = F^*(x_i)$:

$$W^2 = \sum_{i=1}^n \left(z_i - \frac{2i-1}{2n} \right)^2 + \frac{1}{12n}$$

2.3. THE ANDERSON-DARLING (AD) TEST

The Anderson-Darling (AD) test, similarly, is a *quadratic* EDF statistic that can be seen as a modification of CvM. Its defining characteristic is that it assigns even more weight to observations in the tails of the distribution, making it particularly sensitive to discrepancies in these regions. It introduces an additional weighting factor as shown in its formal definition:

$$A^2 = n \int_{-\infty}^{\infty} \frac{(F_n(x) - F^*(x))^2}{F^*(x)(1 - F^*(x))} dF^*(x)$$

The weighting term, $[F^*(x)(1 - F^*(x))]^{-1}$ places greater emphasis on the smallest and largest values of the CDF, a key difference from the KS and CvM tests. For computational purposes from the transformed data points z_i , the statistic is calculated as: (Stephens, 1974)

$$A^2 = - \frac{\sum_{i=1}^n (2i-1)[\ln(z_i) + \ln(1 - z_{n+1-i})]}{n} - n$$

Due to its sensitivity, the AD test is often considered one of the most powerful EDF tests available.

**notes on z_i transformation and uniformity:*

EDF statistics is typically “distribution-free”, when the parameters of the hypothesized distribution are fully specified in advance (Stephens, 1974, “Case 0”). In this case, the probability integral transform $z_i = F(x_i | \mu, \sigma)$ is uniformly distributed, regardless of the original distribution F . Because the test statistics are calculated from these uniform values, their null distributions are independent of F .

When parameters are estimated from the same sample, the transformed values are no longer independent uniform random variables, and standard distribution-free critical values generally do not apply. The present log-normal case, however, is exactly equivalent to the corresponding normal case. If $Y_i = \ln(X_i)$, then

$F_{LN}(X_i | \hat{\mu}, \hat{\sigma}) = \Phi\left(\frac{Y_i - \hat{\mu}}{\hat{\sigma}}\right)$, which is the fitted normal CDF evaluated at Y_i . Therefore, when the same estimators are used, the transformed z_i values—and consequently the KS, CvM, and AD statistics—are identical for the log-normal sample and its log-transformed normal sample. Normal estimated-parameter critical values thus apply directly to the log-normal case under matching estimation conventions. The simulations in this study provide a numerical verification of this equivalence.

3. MONTE CARLO SIMULATION APPROACH

This study uses Monte Carlo simulation to estimate the null distributions of the KS, CvM, and AD statistics and to verify their theoretical equivalence to the corresponding normality-test statistics. The resulting critical values depend on the sample size (n) and the parameter-estimation convention used.

The null (H_0) and alternative (H_1) hypotheses:

H_0 : The sample data are drawn from a log-normal distribution.

H_1 : The sample data are not drawn from a log-normal distribution.

The critical values of interests are **0.75, 0.90, 0.95, and 0.99 quantiles**, corresponding to significance levels $\alpha=0.25, 0.10, 0.05$, and 0.01 , respectively. The simulation was performed for $\mathbf{n} = \{10, 15, 20, 25, 30, 40, 50, 100, 200, 300, 400, 500, 1000, 1500, 2000\}$.

(Razali, 2011) For each sample size, the simulation procedure was executed as follows:

1. Using a fixed random seed of 42, a random sample of size $\mathbf{n}$ was drawn from a standard log-normal distribution (with $\mu=0$ and $\sigma=1$).

2. Transform the observations using $Y_i = \ln(X_i)$ and estimate the parameters by

$$\hat{\mu} = \bar{Y}, \quad \hat{\sigma} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}.$$

Thus, the simulation uses the sample standard deviation with denominator $(n-1)$, rather than the maximum-likelihood estimator with denominator (n) .

3. Sort the transformed observations and calculate $z_i = \Phi\left(\frac{Y_{(i)} - \hat{\mu}}{\hat{\sigma}}\right)$. These are also the

Table 1: Condensed Critical Values for KS, CvM, and AD Tests by Sample Size and Significance Level

Test	Sample Size (n)	$\alpha = 0.25$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
KS	10	0.2093	0.2421	0.2631	0.3037
	50	0.0990	0.1149	0.1248	0.1450
	100	0.0707	0.0816	0.0888	0.1034
	2000	0.0161	0.0186	0.0203	0.0235
CvM	10	0.0719	0.1003	0.1223	0.1716
	50	0.0738	0.1033	0.1263	0.1818
	100	0.0736	0.1041	0.1242	0.1731
	2000	0.0743	0.1043	0.1266	0.1826
AD	10	0.4364	0.5832	0.6993	0.9594
	50	0.4645	0.6329	0.7439	1.0342
	100	0.4679	0.6305	0.7443	0.9950
	2000	0.4704	0.6305	0.7514	1.0434

fitted log-normal CDF values evaluated at the original observations.

4. The KS (D), CvM (W^2), and AD (A^2) statistics were calculated from the z_i values.

5. Steps 1-4 were repeated 10,000 times, creating an empirical null distribution.

6. Estimate each critical value using the corresponding empirical quantile of the 10,000 simulated statistics.

The implementation uses NumPy's default linear quantile interpolation method. This entire process was repeated for each sample size n to generate the final tables.

4. RESULTS AND ANALYSIS

4.1. SIMULATED CRITICAL VALUE TABLES

For KS, CvM, and AD tests for a log-normal null hypothesis with estimated parameters, Table 1 presents a condensed version of these critical values for selected sample sizes. The full, comprehensive tables for all simulated sample sizes are provided in the Appendix (Tables A1, A2, and A3).

The results exhibit different sample-size behavior across the three statistics. The KS critical values decrease at approximately the $n^{-1/2}$ rate. In contrast, the unscaled CvM and AD critical values approach nonzero asymptotic limits rather than decreasing toward zero. Their small nonmonotonic variations across sample sizes reflect Monte Carlo error, particularly for the 0.99 quantile. Because the statistics have different asymptotic scales, their convergence rates should

not be compared directly using the unadjusted critical values.

4.2. VALIDATION: COMPARISON WITH STANDARD BENCHMARKS

To validate the simulation, the estimated critical values are compared with fully specified, distribution-free benchmarks and estimated-normal benchmarks from the literature. The first comparison demonstrates the effect of parameter estimation, while the second verifies the theoretical equivalence between log-normal testing and normality testing of the log-transformed observations.

4.2.1. COMPARISON FOR THE KS TEST

In his comprehensive 1974 paper, Stephens discusses five cases regarding the knowledge of the distribution $F(x)$. For the purpose of this validation, we are interested in two of Stephens' cases (see also *notes on z_i transformation and uniformity*):

Case 0: $F(x)$ is continuous and completely specified.

Case 3: $F(x)$ is the normal distribution with both μ and σ^2 unknown and estimated from the sample.

Specifically, the critical values for the KS test calculated in this study are compared against three standard benchmarks:

- Fully Specified Case (Stephens, 1974, Table

$$D = \text{Constant}/\sqrt{n}$$

1.0): standard, distribution-free (Case 0).

$$\alpha = 0.10 : D = 0.805/\sqrt{50} = 0.114$$

Table 2: Comparison of Log-Normal with Standard Benchmarks for the KS Test (n=50)

Test Case	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
1. Fully Specified Case (Stephens 1.0)	0.170	0.188	0.226
2. Normal Test Case (Stephens 1.3)	0.114	0.125	0.144
3. Normal Test Case (Lilliefors)	0.114	0.125	0.146
4. Log-Normal Test (This Study)	0.115	0.125	0.145
Ratio			
Log-Normal / Fully Specified	0.676	0.665	0.642
Log-Normal / Normal Test (Stephens)	1.009	1.000	1.007
Log-Normal / Normal Test (Lilliefors)	1.009	1.000	0.993

- Normal Test Case (Stephens, 1974, Table 1.3): estimated parameters (Case 3).
- Normal Test Case (Lilliefors, 1967): original Monte Carlo critical values with estimated parameters (Case 3).

Table 2 presents this comparison for a representative sample size of n=50. The benchmark critical values were derived as follows:

1. Fully Specified Case (Stephens, 1974, Table 1.0): reverse-engineered the formula from Stephens' case 0 "*Illustration 1*" (Section 3.3):

$$D = D^*/(\sqrt{n} + 0.12 + 0.11/\sqrt{n})$$

for example,

$$\alpha = 0.10 : D = 1.224/(\sqrt{50} + 0.12 + 0.11/\sqrt{50}) = \mathbf{0.170}$$

2. Estimated Normal Test Case (Stephens, 1974, Table 1.3): reverse-engineered the formula analogous to Stephens' case 3 "*Illustration 2*":

$$D = D^*/(\sqrt{n} - 0.01 + 0.85/\sqrt{n})$$

for example,

$$\alpha = 0.10 : D = 0.819/(\sqrt{50} - 0.01 + 0.85/\sqrt{50}) = \mathbf{0.114}$$

3. Estimated Normal Test Case (Lilliefors, 1967): For n > 30, Lilliefors provides the formula: for example,

These results support the following conclusions:

- The simulated log-normal critical values are approximately **64-68%** of the fully specified values. Thus, using standard distribution-free critical values after estimating the parameters would produce an overly conservative test and reduce power relative to a correctly calibrated test.
- The estimated-normal benchmarks from Stephens (1974) and Lilliefors (1967) are mutually consistent
- The simulated log-normal values closely match the estimated-normal values. This agreement is theoretically expected because applying the fitted log-normal CDF to X_i is equivalent to applying the fitted normal CDF to $Y_i = \ln(X_i)$. Any small discrepancies arise from Monte Carlo variation, rounding, finite-sample corrections, or differences in estimator conventions.

Therefore, normal estimated-parameter KS critical values can be used directly for the log-normal test when the same parameter estimator is used. The CvM and AD comparisons below provide additional numerical checks of this equivalence.

Table 3: Comparison of Log-Normal with Standard Benchmarks for the CvM Test (n=50)

Test Case	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
1. Fully Specified Case (Stephens 1.0)	0.348	0.460	0.736
2. Normal Test Case (Stephens 1.3)	0.103	0.125	0.176
3. Log-Normal Test (This Study)	0.103	0.126	0.182
Ratio			
Log-Normal / Fully Specified	0.296	0.274	0.247
Log-Normal / Normal Test	1.000	1.008	1.034

Table 4: Comparison of Log-Normal with Standard Benchmarks for the AD Test (n=50)

Test Case	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
1. Fully Specified Case (Stephens 1.0)	1.933	2.492	3.857
2. Normal Test Case (Stephens 1.3)	0.613	0.736	1.021
3. Log-Normal Test (This Study)	0.633	0.744	1.034
Ratio			
Log-Normal / Fully Specified	0.327	0.299	0.268
Log-Normal / Normal Test	1.033	1.011	1.013

4.2.2. COMPARISON FOR THE CVM TEST

The same comparison was performed for the CvM (W^2) statistic. Table 3 compares the simulated critical values for the log-normal test against the standard benchmarks for the fully specified case (Case 0) and the normal test with estimated parameters (Case 3) from Stephens (1974).

Similar to the KS case, the benchmark critical values for n=50 were derived as follows:

1. Fully Specified Case (Stephens, 1974, Table 1.0): W^2 is calculated by algebraically solving Stephens' modification formula: $W^{2*} = (W^2 - 0.4/n + 0.6/n^2)(1.0 + 1.0/n)$.

for example,

$\alpha = 0.10$: $W^2 = (0.347/1.02) + 0.00776 = \mathbf{0.348}$

2. Estimated Normal Test Case (Stephens, 1974, Table 1.3): W^2 is found by reverse-engineering the formula for Case 3: $W^{2*} = W^2(1.0 + 0.5/n)$.

for example,

$\alpha = 0.10$: $W^2 = 0.104/1.01 = \mathbf{0.103}$

The CvM comparison supports two conclusions:

- The simulated critical values are only about **25–30%** of the fully specified, distribution-free values. Therefore, applying fully specified CvM critical values after estimating the parameters would produce a substantially conservative test.
- The simulated log-normal values closely match the estimated-normal values. As with the KS statistic, this agreement is theoretically required when the same estimators are used because both tests are calculated from identical fitted CDF values. Small differences reflect Monte Carlo variation, rounding, or differences in finite-sample and estimator conventions.

Thus, estimated-normal CvM critical values apply directly to the corresponding log-normal test under matching estimation conventions. The comparison at (n=50) provides a numerical verification of this equivalence rather than evidence limited to small sample sizes.

4.2.3. COMPARISON FOR THE AD TEST

Finally, the same comparison was performed for the Anderson-Darling (A^2) statistic. Table 4 compares the simulated critical values for the log-normal test against the standard benchmarks for the fully specified case (Case 0) and the normal test with estimated parameters (Case 3) from Stephens (1974)

The benchmark critical values for n=50 were derived as follows:

1. Fully Specified Case (Stephens, 1974, Table 1.0): Stephens notes that for A^2 , no modification is required for $n \geq 5$. Therefore, the sample-size-specific critical value is the same as the published asymptotic value. For example, $\alpha = 0.10$: $A^2 = \mathbf{1.933}$

2. Estimated Normal Test Case (Stephens, 1974, Table 1.3): A^2 is found by reverse-engineering the formula for Case 3:

$$A^{2*} = A^2(1.0 + 4/n - 25/n^2)$$

for example,

$\alpha = 0.10$: $A^2 = 0.656/1.07 = \mathbf{0.613}$

The AD comparison produces the same general conclusions:

- The simulated critical values are approximately **27–33%** of the fully specified, distribution-free values. Consequently, using fully specified AD critical values after estimating the parameters would result in a severely conservative test.
- The simulated log-normal values closely match the estimated-normal values. Although the AD

between log-normal goodness-of-fit testing and normality testing of the log-transformed observations. They should not be interpreted as evidence that parameter estimation generally makes EDF null distributions insensitive to the underlying distribution family.

5. CONCLUSION AND FUTURE WORK:

This study estimated finite-sample critical values for the KS, CvM, and AD goodness-of-fit tests for a log-normal distribution with parameters estimated on the log scale. The simulations produced values closely matching established estimated-normal benchmarks. This agreement follows from an exact theoretical equivalence: fitting a log-normal distribution to X_i is the same as fitting a normal distribution to $Y_i = \ln(X_i)$, so both procedures produce identical fitted CDF values and EDF statistics when the same estimators are used.

The results also demonstrate that fully specified, distribution-free critical values are inappropriate when parameters are estimated from the sample. Such values are substantially larger and therefore produce overly conservative tests. In contrast, estimated-normal critical values can be applied directly to log-normal goodness-of-fit tests under matching parameter-estimation conventions.

Future work could increase the number of replications and report Monte Carlo confidence intervals, particularly for the 0.99 quantiles. The effects of alternative parameter estimators could also be examined because changing the estimator changes the null distribution. Finally, similar analyses could be conducted for other distribution families, but equivalence with normal critical values should not be assumed unless it can be established theoretically. Power comparisons against selected alternative distributions would provide an additional assessment of the practical performance of the three tests.

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(FUN NOTE ON COMPUTATIONAL PROGRESS)

Lewis, in his foundational 1961 paper on the distribution of the Cramér-von Mises statistic, remarked that determining the full distribution via Monte Carlo methods was "*prohibitive*." He noted that for a sample size of just $n=8$, a simulation of 250,000 replications required **six hours** of computing time on the advanced machines of the era, an IBM 704 Computer

With the computing power nowadays, this can be replicated in probably minutes on a standard laptop running a language like Python, with languages like C++ or Fortran likely even faster. This is not meant to diminish the groundbreaking efforts of these pioneers! On the contrary, it highlights the immense challenges they overcame not too long ago. For contrast, the entire simulation for this paper—including 3 tests, 15 sample sizes (ranging from $n=10$ to 2000), and 10,000 replications for each—was executed on a standard modern laptop using Python in 8.06 seconds, so with 250,000 replications, likely it will take less than 4 minutes.

Table A1. Full Critical Values for the Kolmogorov-Smirnov Test (D)

Sample Size (n)	$\alpha = 0.25$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
10	0.2093	0.2421	0.2631	0.3037
15	0.1739	0.2014	0.2187	0.2513
20	0.1514	0.1748	0.1910	0.2219
25	0.1377	0.1592	0.1733	0.2015
30	0.1254	0.1454	0.1582	0.1834
40	0.1100	0.1271	0.1384	0.1592
50	0.0990	0.1149	0.1248	0.1450
100	0.0707	0.0816	0.0888	0.1034
200	0.0507	0.0584	0.0634	0.0740
300	0.0412	0.0477	0.0520	0.0605
400	0.0356	0.0410	0.0445	0.0526
500	0.0320	0.0371	0.0403	0.0470
1000	0.0228	0.0263	0.0286	0.0332
1500	0.0185	0.0213	0.0231	0.0268
2000	0.0161	0.0186	0.0203	0.0235

statistic places greater weight on the distribution tails, it is calculated from the same fitted CDF values in both cases. Its equivalence therefore follows from the logarithmic transformation and matching parameter estimators, just as it does for the KS and CvM statistics.

The consistent results across all three tests numerically confirm the theoretical equivalence

Table A2. Full Critical Values for the Cramér-von Mises Test (W^2)

Sample Size (n)	$\alpha = 0.25$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
10	0.0719	0.1003	0.1223	0.1716
15	0.0728	0.1013	0.1220	0.1670
20	0.0719	0.0995	0.1200	0.1682
25	0.0731	0.1014	0.1240	0.1707
30	0.0723	0.1005	0.1223	0.1754
40	0.0732	0.1030	0.1238	0.1727
50	0.0738	0.1033	0.1263	0.1818
100	0.0736	0.1041	0.1242	0.1731
200	0.0736	0.1040	0.1254	0.1796
300	0.0732	0.1042	0.1264	0.1841
400	0.0738	0.1017	0.1224	0.1753
500	0.0741	0.1034	0.1244	0.1798
1000	0.0742	0.1043	0.1263	0.1802
1500	0.0740	0.1017	0.1220	0.1766
2000	0.0743	0.1043	0.1266	0.1826

Table A3. Full Critical Values for the Anderson Darling Test (A^2)

Sample Size (n)	$\alpha = 0.25$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
10	0.4364	0.5832	0.6993	0.9594
15	0.4498	0.6015	0.7095	0.9464
20	0.4486	0.5986	0.7085	0.9599
25	0.4579	0.6109	0.7339	0.9828
30	0.4551	0.6050	0.7224	1.0024
40	0.4605	0.6161	0.7322	1.0004
50	0.4645	0.6329	0.7439	1.0342
100	0.4679	0.6305	0.7443	0.9950
200	0.4731	0.6281	0.7470	1.0168
300	0.4648	0.6326	0.7609	1.0609
400	0.4666	0.6204	0.7301	1.0186
500	0.4701	0.6241	0.7417	1.0224
1000	0.4746	0.6358	0.7489	1.0509
1500	0.4687	0.6189	0.7360	1.0154
2000	0.4704	0.6305	0.7514	1.0434